

Element I: Testing, Data Collection, and Analysis

For our school rain garden project, we focused on creating detailed designs and plans to manage stormwater through natural filtration and retention. With support from college students at UMD, we researched soil types, plant choices, and drainage solutions to make sure our design met environmental goals. While we didn't conduct physical tests, we used modeling and data from similar projects to estimate things like infiltration rates and water flow. This helped us make informed design decisions and understand how our rain garden would perform in real conditions. Working with the UMD students made the planning process more insightful and helped us think critically about how to turn our ideas into an effective, sustainable solution.